

Budget Justification: DOE Genesis Mission
(07/01/2026 - 03/31/2027)

Project Title: Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

A. Senior Personnel

	Salary
PI: Mark Sussman (1.0 mo)	\$16,188
Co-PI: Kourosh Shoele (1.0 mo)	\$15,146

Salary for one summer month is requested for the PI, Dr. Sussman, and Co-PI, Dr. Shoele, to lead the research, develop the AI-assisted multiphase models, and supervise the students and postdoc.

Mark Sussman: Monthly salary is based on a current 9-month salary of \$143,480.16 (\$16,187.50 per month).

Kourosh Shoele: Monthly salary is based on a current 9-month salary of \$134,249.05 (\$15,146.00 per month).

Senior Personnel Fringe Benefits.

	Year 1
PI and Co-PI (35%)	\$10,967

Fringe benefits are calculated based on FSU's standard rates for faculty (35%).

Total Senior Personnel: \$42,301.

B. Other Personnel

	Salary (9 months)
1 Postdoctoral Researcher (1.0 FTE)	\$48,750
2 Graduate Students (0.5 FTE)	\$46,579

Salaries are requested to support one postdoctoral researcher (1.0 FTE for 9 months) and two graduate students (0.5 FTE each). They will work with the PI and Co-PI on the proposed research.

- Postdoctoral Researcher: One postdoc will be supported for 9 months to work on the implementation of high-fidelity simulations and AI training. The salary is calculated based on an annual rate of \$65,000, totaling \$48,750 for the 9-month period.
- Graduate Students: Two Ph.D. students will be appointed at 0.5 FTE (20 hours/week). Following FSU Department of Mathematics guidelines, the stipend is set at \$11,644.63 per student per semester (Fall and Spring), totaling \$46,578.52 for the grant period.

Other Personnel Fringe Benefits

	Year 1
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Postdoc (27.7%)	\$13,504
Graduate Students (6.4%)	\$2,981

Total Other Personnel: \$111,814

Total Salary, Wages and Fringe Benefits: \$154,115

C. Equipment Description

No capital equipment exceeding \$10,000 is requested for this computational study. Research will utilize existing workstation infrastructure and the FSU High Performance Computing (HPC) cluster.

D. Travel

	Year 1
Domestic	\$7,000
International	\$4,500

Domestic Travel: Funds are requested for the PI and Co-PI to attend the APS Division of Fluid Dynamics (DFD) Annual Meeting and the DOE Genesis Mission PI meeting. Costs cover registration, airfare, and per diem for two trips.

International Travel: Funds (\$4,500) are requested for one team member to present findings at the International Conference on Liquid Atomization and Spray Systems (ICLASS). This includes international airfare, registration, and lodging/meals at the FSU estimated rate.

- Domestic Travel (\$7,000): Supports the PI and Co-PI to attend the APS DFD Annual Meeting and the DOE Genesis Mission PI meeting (registration, airfare, and per diem).
- International Travel (\$4,500): Supports one team member to present project findings at ICLASS (international airfare, registration, lodging, and meals).

Total Travel Costs: \$11,500.

F. Other Direct Costs

1. Materials & Supplies: A budget of \$5,000 is requested for research-related expenses, including departmental computer access charges (\$235/computer including AppleCare) and specialized software licenses required for CFD post-processing (e.g., Tecplot 360 or MATLAB).

4. Computer Services: \$11,932 is requested for FSU High Performance Computing (HPC) services. This includes 1 million NCU-hours for training the AI models and running high-fidelity multiphase flow simulations (\$10,000) and 21TB of high-speed storage for simulation data (\$1,932).

8. Tuition: Tuition for two graduate students is requested per FSU's standard rates for 18 credit hours over the grant period totals \$14,672.

Total Other Direct Cost: \$31,604

G. Direct Costs

Total Direct Costs (A thru F) \$197,219

H. Indirect Costs

Indirect costs are calculated at the FSU rate of 54% of the Modified Total Direct Costs (MTDC). MTDC excludes student tuition and capital equipment. Included below is the current facilities and administrative (F&A) cost Rate Agreement.

Total Indirect Costs: 98,575

I. Total Direct and Indirect Costs

Total Direct and Indirect Institutional Costs: \$295,794



DEPARTMENT OF HEALTH & HUMAN SERVICES

Program Support Center
Financial Management Portfolio
Cost Allocation Services

5600 Fishers Lane | Office 08N144
Rockville, MD 20857
EMAIL: CAS-Bethesda@psc.hhs.gov

June 27, 2024

Stacey Patterson
Vice President for Research
Florida State University
214 Westcott Building
Tallahassee, FL 32306

Dear Ms. Patterson:

A copy of the facilities and administrative (F&A) cost Rate Agreement is being mailed to you for your signature. This agreement reflects an understanding reached between your organization and a member of my staff concerning the rate(s) that may be used to support your claim for F&A and fringe benefit costs on grants and contracts with the Federal Government.

Please have the agreement signed by an authorized representative of your organization and send to me, retaining a copy for your files. We will reproduce and distribute the Rate Agreement to the appropriate awarding organizations of the Federal Government for their use.

A fringe benefit cost proposal, together with the required supporting information, must be submitted to this office for each fiscal year in which your organization claims fringe benefits under grants and contracts awarded by the Federal Government. Therefore, your next fringe benefit proposal for the fiscal year ending June 30, 2024 will be due in our office by December 31, 2024. The proposals and supporting documents should be emailed to CAS-Bethesda@psc.hhs.gov.

Sincerely,

Darryl W. Mayes
Deputy Director
Cost Allocation Services

CONCURRENCE:

COLLEGES AND UNIVERSITIES RATE AGREEMENT

EIN: 1591961248A3
ORGANIZATION:
Florida State University
Office – Sr. V.P. for Fin. & Admin.
214 Westcott Building
Tallahassee, FL 32306-1320

Date: 06/27/2024
FILING REF.: The preceding
agreement was dated
01/25/2024

The rates approved in this agreement are for use on grants, contracts and other agreements with the Federal Government, subject to the conditions in Section III.

SECTION I: INDIRECT COST RATES

RATE TYPES:		FIXED	FINAL	PROV. (PROVISIONAL)	PRED. (PREDETERMINED)
<u>EFFECTIVE PERIOD</u>					
<u>TYPE</u>	<u>FROM</u>	<u>TO</u>	<u>RATE(%)</u>	<u>LOCATION</u>	<u>APPLICABLE TO</u>
PRED.	07/01/2023	06/30/2025	54.00	On-Campus	Organized Research (1)
PRED.	07/01/2023	06/30/2025	26.00	Off-Campus	Organized Research (1)
PRED.	07/01/2023	06/30/2025	50.00	On-Campus	Instruction (1)
PRED.	07/01/2023	06/30/2025	26.00	Off-Campus	Instruction (1)
PRED.	07/01/2023	06/30/2025	45.00	On-Campus	Other Sponsored Activity (1)
PRED.	07/01/2023	06/30/2025	26.00	Off-Campus	Other Sponsored Activity (1)
PRED.	07/01/2023	06/30/2025	58.50	On-Campus	NHMFL (2)
PRED.	07/01/2023	06/30/2025	26.00	Off-Campus	NHMFL (2)
PROV.	07/01/2025	Until Amended			Use same rates and conditions as those cited for fiscal year ending June 30, 2025.

*BASE

Modified total direct costs, consisting of all direct salaries and wages, applicable fringe benefits, materials and supplies, services, travel and up to the first \$25,000 of each subaward (regardless of the period of performance of the subawards under the award). Modified total direct costs shall exclude equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs and the portion of each subaward in excess of \$25,000. Other items may only be excluded when necessary to avoid a serious inequity in the distribution of indirect costs, and with the approval of the cognizant agency for indirect costs.

SECTION I: FRINGE BENEFIT RATES**

TYPE	FROM	TO	RATE(%)	LOCATION	APPLICABLE TO
FIXED	7/1/2023	6/30/2024	2.10	All	All Staff (1)
FIXED	7/1/2023	6/30/2024	0.10	All	All Staff (2)
FIXED	7/1/2024	6/30/2025	27.90	All	Clinical Faculty
FIXED	7/1/2024	6/30/2025	30.30	All	Faculty / Exec Svc
FIXED	7/1/2024	6/30/2025	41.40	All	A&P
FIXED	7/1/2024	6/30/2025	50.70	All	Charter Sch Faculty
FIXED	7/1/2024	6/30/2025	54.30	All	USPS
FIXED	7/1/2024	6/30/2025	8.40	All	OPS & Temps / GAs
FIXED	7/1/2024	6/30/2025	21.00	All	Post-Docs
FIXED	7/1/2024	6/30/2025	1.70	All	Students
FIXED	7/1/2024	6/30/2025	2.00	All	Athletic Coach Suppl
FIXED	7/1/2024	6/30/2025	6.20	All	Bonus / One-Time (A)
FIXED	7/1/2024	6/30/2025	18.50	All	Overtime / Suppl (B)
PROV.	7/1/2025	Until Amended			Use same rates and conditions as those cited for fiscal year ending June 30, 2025

**** DESCRIPTION OF FRINGE BENEFITS RATE BASE:**

Salaries and wages.

- (1) Applicable to all faculty and staff excluding temporary employees.
(2) Applicable to all faculty and staff including temporary employees.

SECTION II: SPECIAL REMARKS

TREATMENT OF FRINGE BENEFITS:

Effective up to and including 06/30/2024:

For cash claims and final reporting purposes, the following benefits are specifically identified to each employee and charged individually as direct costs: FICA, Retirement, Life, Pre-tax Assessments, and Health Insurance.

Employee Terminal Leave, Worker's Compensation, Paid Parental Leave, and Unemployment costs are charged using the rate(s) listed in Section I of this Rate Agreement.

Effective 07/01/2024:

The fringe benefits are charged using the rate(s) listed in the Fringe Benefits Section of this Agreement. The fringe benefits included in the rate(s) are Health Insurance, Retirement, Life and Disability Insurance, Workers Compensation, Unemployment Insurance, FICA, FICA Medicare, Terminating Leave Payout, and Paid Parental Leave.

TREATMENT OF PAID ABSENCES:

Vacation, holiday, sick leave pay and other paid absences are included in salaries and wages and are claimed on grants, contracts and other agreements as part of the normal cost for salaries and wages. Separate claims are not made for the cost of these paid absences.

OFF-CAMPUS DEFINITION: The off-campus rate will apply for all activities: a) Performed in facilities not owned by the institution and where these facility costs are not included in the F&A pools; or b) Where rent is directly allocated/charged to the project(s). Grants or contracts will not be subject to more than one F&A cost rate. If more than 50% of a project is performed off-campus, the off-campus rate will apply to the entire project.

Equipment means an article of nonexpendable tangible personal property having a useful life of more than one year, and an acquisition cost of \$5,000 or more per unit.

Indirect cost rate and fringe benefit rate proposals based on the fiscal year ending June 30, 2024 are due in our office by December 31, 2024.

SECTION III: GENERAL

A. LIMITATIONS:

The rates in this Agreement are subject to any statutory or administrative limitations and apply to a given grant, contract or other agreement only to the extent that funds are available. Acceptance of the rates is subject to the following conditions: (1) Only costs incurred by the organization were included in its indirect cost pool as finally accepted: such costs are legal obligations of the organization and are allowable under the governing cost principles; (2) The same costs that have been treated as indirect costs are not claimed as direct costs; (3) Similar types of costs have been accorded consistent accounting treatment; and (4) The information provided by the organization which was used to establish the rates is not later found to be materially incomplete or inaccurate by the Federal Government. In such situations the rate(s) would be subject to renegotiation at the discretion of the Federal Government.

B. ACCOUNTING CHANGES:

This Agreement is based on the accounting system purported by the organization to be in effect during the Agreement period. Changes to the method of accounting for costs which affect the amount of reimbursement resulting from the use of this Agreement require prior approval of the authorized representative of the cognizant agency. Such changes include, but are not limited to, changes in the charging of a particular type of cost from indirect to direct. Failure to obtain approval may result in cost disallowances.

C. FIXED RATES:

If a fixed rate is in this Agreement, it is based on an estimate of the costs for the period covered by the rate. When the actual costs for this period are determined, an adjustment will be made to a rate of a future year(s) to compensate for the difference between the costs used to establish the fixed rate and actual costs.

D. USE BY OTHER FEDERAL AGENCIES:

The rates in this Agreement were approved in accordance with the authority in Title 2 of the Code of Federal Regulations, Part 200 (2 CFR 200), and should be applied to grants, contracts and other agreements covered by 2 CFR 200, subject to any limitations in A above. The organization may provide copies of the Agreement to other Federal Agencies to give them early notification of the Agreement.

E. OTHER:

If any Federal contract, grant or other agreement is reimbursing indirect costs by a means other than the approved rate(s) in this Agreement, the organization should (1) credit such costs to the affected programs, and (2) apply the approved rate(s) to the appropriate base to identify the proper amount of indirect costs allocable to these programs.

BY THE INSTITUTION:

Florida State University

(INSTITUTION)



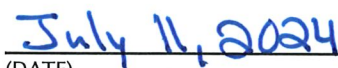
(SIGNATURE)



(NAME)



(TITLE)



(DATE)

ON BEHALF OF THE GOVERNMENT:DEPARTMENT OF HEALTH AND HUMAN SERVICES

(AGENCY)

-S

Digitally signed by Darryl W.

Mayes -S

Date: 2024.06.28 14:54:00 -04'00'

(SIGNATURE)

Darryl W. Mayes

(NAME)

Deputy Director, Cost Allocation Services

(TITLE)

06/27/2024

(DATE)

HHS REPRESENTATIVE: Steven ZurafTELEPHONE: (301) 492-4855